

# Austin Tan

[tan.austin0903@gmail.com](mailto:tan.austin0903@gmail.com) • 929-406-7929 • [linkedin.com/in/austin-tan903](https://www.linkedin.com/in/austin-tan903) • [github.com/austintan903](https://github.com/austintan903) • [austintan/me](https://austintan.me)

## EDUCATION

### New York University (NYU)

*Bachelor of Science in Computer Science (GPA: 3.9/4.0)*

- **Relevant Coursework:** Data Structures and Algorithms, Design and Analysis of Algorithms, Operating Systems, Machine Learning, Artificial Intelligence, Computer Architecture, Databases, Probability & Statistics, Discrete Mathematics
- **Activities:** Tech@NYU Dev Team - Spring 2025

New York City, NY

*Expected December 2027*

## TECHNICAL SKILLS

**Languages:** C++, Python, Typescript, Javascript, SQL, HTML, CSS, Bash

**Frameworks & Libraries:** React.js, Next.js, Falcon, Node.js, Pandas, NumPy, scikit-learn, Matplotlib, SQLAlchemy

**Developer Tools:** Docker, Kubernetes, PostgreSQL, Redis, CMake, GoogleTest, GCP, AWS, Git, Linux, Unix, CI/CD, Vercel, Jenkins

**Testing & Practices:** Unit Testing, Integration Testing, Test-Driven Development

**AI Tools:** Claude Code, Cursor, Codex

## WORK EXPERIENCE

### MathWorks

*Incoming Engineering Development Group Intern*

- MATLAB Language QE Team

Natick, MA

*Summer 2026*

### New York University

*Undergraduate Research Assistant*

- Developing a novel WAN consensus protocol in C++ under Prof. Jinkun Geng that eliminates coordination and enabling independent request processing across replicas
- Deployed multi-region GCP VMs (US, Europe, Asia, South America) and engineered low-level networking to measure WAN latency, outperforming existing protocols (VR, Nezha, Multi-Paxos) in throughput and latency

New York City, NY

*Apr 2026 - Present*

### CrowdVolt (YC W24)

*Software Engineering Intern*

- Designed and implemented a price alert feature used 4000+ times across the marketplace; designed PostgreSQL database schema; built REST APIs and delivered Next.js/React frontend; supported end-to-end production roll out, monitoring, and reliability
- Built an AI-powered event request aggregation pipeline in Python using LLMs to normalize 1,000+ requests across artist, location, and date; surfaced results through an internal operations dashboard reducing time for team triage and review
- Developed cart validation logic to detect and block marketplace manipulation instances to near 0
- Built a Redis-backed connection error triage service across a multi-stage bank verification pipeline, increasing seller bank linking conversion; included admin tooling to unblock stalled verifications and reduce manual ops intervention
- Fully developed and designed an internal version of Jira used daily by 15+ company employees; also developed a customer support shift scheduling tool, price alert analytics dashboard, and banned user transaction monitor for the operations team

New York City, NY

*Oct 2025 - Apr 2026*

## SELECTED PROJECTS

**Distributed Append Only Log** | C++, liburing, Linux, Docker, CMake, GoogleTest

- Built a high-performance distributed append only log in C++20; designed a custom binary wire protocol with packed structs, network byte-order encoding, and sequence-numbered UDP ACKs with producer-side timeout/retransmit; built a CLI client for producing and consuming messages across topics in a multi-node network
- Implemented the persistence layer using io\_uring for high-throughput async I/O; stored records in a CRC32 validated binary format with crash recovery and partial write truncation; ensured multi-thread safety across concurrent writers; verified correctness with 40+ unit and concurrency tests via Google Test
- Scaled the system horizontally by partitioning topics across 3 independent replica groups; implemented Network-Ordered Paxos (NOPaxos) with a sequencer to stamp total order on log entries, enabling replication without replica coordination, achieving 0.53 ms median latency and 1,900 ops/sec sustained throughput; built a health monitor that tracks node liveness and triggers automatic failover

**Machine Learning Music Recommendation System** | Python, NumPy, Pandas, scikit-learn, Matplotlib, Seaborn, Gradio

- Engineered a hybrid recommender combining SVD (optimized with SGD for 20 epochs) and KNN, improving Precision@5 by 150% (0.036→0.091) and reducing training MSE by 60% (1.08→0.43)
- Processed 166K+ Last.fm interactions across 76K+ songs, matching 114K+ Spotify tracks; extracted 13 audio features and applied Z-score normalization and cosine similarity for content modeling
- Built evaluation framework comparing SVD, KNN, and hybrid models using Precision/Recall/NDCG@k (5,10,20) on a 133K+ interaction dataset (75/12.5/12.5 split) across an 84% sparse user-item matrix; generated t-SNE (5K embeddings), feature correlation heatmaps, and convergence curves

**ClassroomLM (Tech@NYU Dev Team Spring 25)** | Typescript, Next.js, React.js, Supabase, RAG, shadcn/ui

- Co-developed an open-source classroom LLM assistant using RAGFlow and Supabase that grounds responses in per-classroom document knowledge bases for improved retrieval accuracy
- Built embedding pipelines for PDFs, slides, and handwritten notes into RAGFlow knowledge bases, enabling context-specific responses for niche course material
- Implemented real-time collaborative study chats via Supabase Realtime where an AI participates as a full conversation member with persistent context via /ask commands